Window function

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Lab 1: Film Ranking (The Basics)

```sql
--Lab 01:
with Rankedfilms as(
    select c.name as category, f.title as film_title, f.length as film_len,
    rank() over (partition by c.name order by f.length desc, f.title) as rn
    from film f
    join film_category fc on f.film_id = fc.film_id
    join category c on c.category_id = fc.category_id
)
select * from Rankedfilms where rn <= 3
```

	category character varying (25)	film_title character varying (255)	film_len smallint	rn bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Name	182	2
15	Comedy	Searchers Wait	182	3
16	Documentary	Wife Turn	183	1
17	Documentary	Young Language	183	2
18	Documentary	Cause Date	179	3
19	Drama	Jacket Frisco	181	1
20	Drama	Something Duck	180	2

Lab 2: GrowthAnalysis (Time Series)

```
--Lab 02:
WITH monthly_revenue AS (
    SELECT
        DATE_TRUNC('month', payment_date)::date AS revenue_month,
        SUM(amount) AS total_revenue
    FROM payment
    GROUP BY DATE_TRUNC('month', payment_date)
),
revenue_with_lag AS (
    SELECT
        revenue_month,
        total_revenue,
        LAG(total_revenue) OVER (ORDER BY revenue_month) AS previous_revenue
    FROM monthly_revenue
)
SELECT
    revenue_month,
    total_revenue,
    previous_revenue,
    ROUND(
        (total_revenue - previous_revenue) / previous_revenue * 100,
        2
    ) AS mom_growth_pct
FROM revenue_with_lag
ORDER BY revenue_month;
```

	revenue_month date	total_revenue numeric	previous_revenue numeric	mom_growth_pct numeric
1	2007-02-01	8351.84	[null]	[null]
2	2007-03-01	23886.56	8351.84	186.00
3	2007-04-01	28559.46	23886.56	19.56
4	2007-05-01	514.18	28559.46	-98.20

Lab 3: CustomerLifetime Value (Running Total)

```sql
-- Lab 3:
SELECT
    customer_id,
    payment_id,
    payment_date,
    amount,
    SUM(amount) OVER (
        PARTITION BY customer_id
        ORDER BY payment_date
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS running_clv
FROM payment
ORDER BY customer_id, payment_date;
```

	customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	running_clv numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92
9	1	22681	2007-03-02 14:05:18.996577	3.99	40.91
10	1	22682	2007-03-02 16:30:04.996577	0.99	41.90
11	1	22683	2007-03-17 11:06:20.996577	4.99	46.89
12	1	22684	2007-03-18 02:25:55.996577	0.99	47.88
13	1	22685	2007-03-19 08:23:42.996577	0.99	48.87
14	1	22686	2007-03-19 12:25:20.996577	2.99	51.86
15	1	22687	2007-03-21 22:02:23.996577	0.99	52.85
16	1	22688	2007-03-21 23:56:23.996577	1.99	54.84
17	1	22689	2007-03-22 18:10:03.996577	2.99	57.83
18	1	22690	2007-03-22 18:32:12.996577	5.99	63.82
19	1	28993	2007-04-08 01:45:31.996577	5.99	69.81
20	1	28994	2007-04-08 06:02:22.996577	5.99	75.80

Lab 4: The”Top-N”Filter Challenge

```sql
-- Lab 4:
WITH customer_spend_by_country AS (
    SELECT
        cu.customer_id,
        cu.first_name,
        cu.last_name,
        co.country,
        SUM(p.amount) AS total_spend
    FROM customer cu
    JOIN address a ON cu.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON ci.country_id = co.country_id
    JOIN payment p ON cu.customer_id = p.customer_id
    GROUP BY cu.customer_id, cu.first_name, cu.last_name, co.country
),
ranked_customers AS (
    SELECT
        customer_id,
        first_name,
        last_name,
        country,
        total_spend,
        RANK() OVER (
            PARTITION BY country
            ORDER BY total_spend DESC
        ) AS spend_rank
    FROM customer_spend_by_country
)
SELECT
```

```sql
SELECT
    country,
    customer_id,
    first_name,
    last_name,
    total_spend
FROM ranked_customers
WHERE spend_rank = 1
ORDER BY country;
```

	country character varying (50)	customer_id integer	first_name character varying (45)	last_name character varying (45)	total_spend numeric
1	Afghanistan	218	Vera	Mccoy	67.82
2	Algeria	176	June	Carroll	151.68
3	American Samoa	320	Anthony	Schwab	47.85
4	Angola	528	Claude	Herzog	93.80
5	Anguilla	381	Bobby	Boudreau	99.68
6	Argentina	560	Jordan	Archuleta	129.71
7	Armenia	41	Stephanie	Mitchell	118.75
8	Austria	266	Nora	Herrera	113.74
9	Azerbaijan	334	Raymond	Mcwhorter	108.75
10	Bahrain	590	Seth	Hannon	108.76
11	Bangladesh	21	Michelle	Clark	146.68
12	Belarus	144	Clara	Shaw	189.60
13	Bolivia	431	Joel	Francisco	92.78
14	Brazil	178	Marion	Snyder	194.61
15	Brunei	91	Lois	Butler	107.66
16	Bulgaria	540	Tyrone	Asher	112.76
17	Cambodia	516	Elmer	Noe	92.76
18	Cameroon	352	Albert	Crouse	96.78
19	Canada	410	Curtis	Irby	167.62
20	Chad	535	Javier	Elrod	122.72